An Oriented Object Detector for Hazy Remote Sensing Images

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***Abstract*— Currently, a lot of work is focused on aerial object detection and has achieved good results. Though these methods have achieved promising results on the conventional datasets, it is still challenging to locate objects from the low-quality images captured in adverse weather conditions. Currently, there are limited approaches that combine aerial object detection with hazy conditions, and there are few publicly available datasets for real hazy weather based on aerial images. For this purpose, we propose a dataset HRSI, hazy remote sensing images in the real world, which is mainly divided into three categories: airport, large vehicle, and ship. All images in HRSI are from real hazy conditions. In addition, we propose an object detection model DFENet, a dehazing feature enhancement model for hazy remote sensing images, which is suitable for hazy weather. DFENet consists of a two-branch and a dehazing module. The two-branch structure helps to fully learn hazy and dehazing features. In order to avoid the impact of noise caused by the dehezing module, we also designed a haze-predict module (HPM) to predict the information containing haze in the image. We introduce the cross-fuse module (CFM) to utilize the information of haze to guide the feature fusion of two branches. By utilizing the information of haze, DFENet can dynamically adjust the feature weight in the two-branch to avoid the impact of noise generated by the dehazing module. Compared with traditional object detection methods, DFENet not only has good performance in hazy conditions but also improves performance in clear conditions. We tested DFENet on DOTA, HRSI, and Foggy-DOTA to demonstrate that DFENet performs better under hazy conditions.**

***Index Terms*— Aerial object detection, hazy object detection, weakly supervised.**

I. Introduction

With the development of deep learning, object detection methods based on deep neural networks have replaced traditional methods and become the mainstream. These methods typically have higher accuracy and better performance, and require less manual design when implemented. Universal object detection models such as [1], [2], [3], [4], [5], and [6], have achieved significant success in natural images, but it is challenging to directly transfer deep learning-based object detection methods to optical remote sensing images. Remote sensing object detection is different from general object detection tasks. Remote sensing object localization has the following difficulties: nonaxis-aligned objects with arbitrary directions and dense objects with complex backgrounds. Most improved methods [7], [8], [9], [10] predict frames with rotation angles to replace horizontal frames. Although these methods have achieved amazing results in remote sensing object detection. Despite advancements in the field, the detection performance of remote sensing object detection remains suboptimal under degraded conditions, such as those encountered during hazy, rainy, or low-light periods. This limitation stems from the propensity of deep learning models to overfit to the training datasets, which results in diminished performance when confronted with test data that includes degraded imagery. The challenge of maintaining robust detection capabilities in the presence of such conditions continues to be a significant issue within the domain of remote sensing.

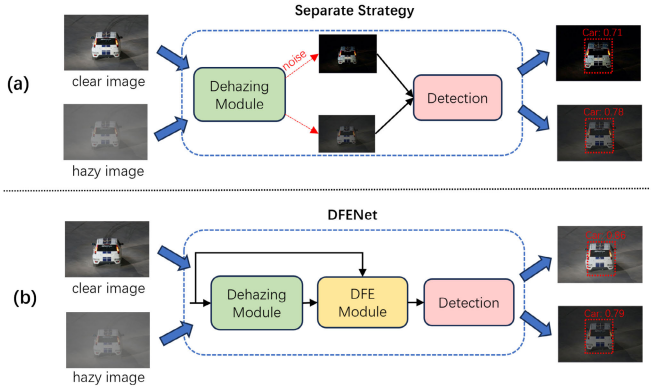


Fig. 1. Performance of the dehazing module in different detection models. (a) Image goes through the dehazing module before being detected. When clean images are input into the dehazing module, noise is introduced, leading to a decrease in the performance of the detection module. The input of hazy images into the dehazing module will make the images clearer and improve the performance of the detection module. (b) Our proposed DFENet. Our model has a dehazing feature enhancement module (DFE), and when a clean image is input, the DFE will impose a penalty on the dehazing image, making the output feature more inclined toward a clean feature. When inputting a hazy image, the DFE will make the output feature more inclined toward the feature after dehazing.

As shown in Fig. 1, for object detection of degraded images, many traditional methods first train a dehazing module to restore the image, and then directly input the dehazed image into the network for training. These dehazing modules generally use generative models, such as [11], [12], [13], and

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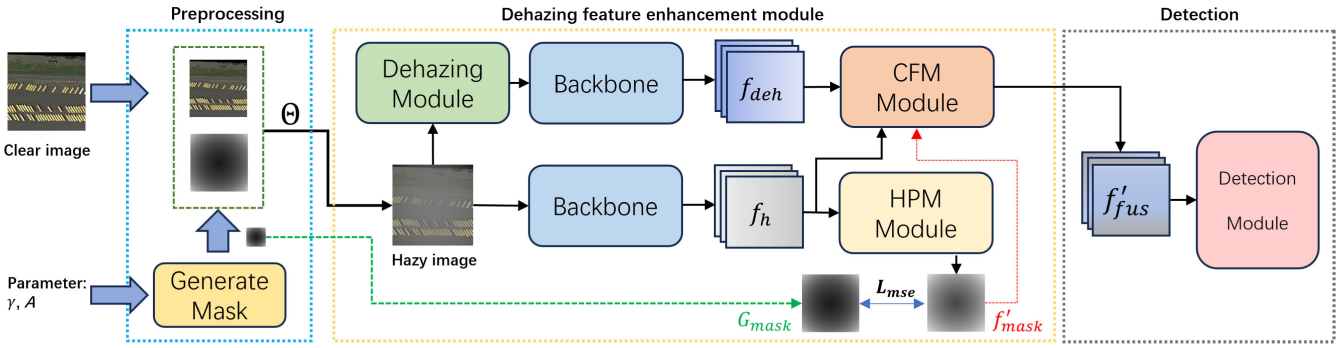


Fig. 2. Architecture of proposed DFENet. The blue dotted line represents the process of dehazing, the yellow dotted line represents the DFE, and the black dotted line represents the detection module. In the process of Algorithm 1, clean image and some parameters are input, then we get the G_{mask} and the haze image used for training, which will be used during training, Θ is the composite function defined in Algorithm 1. We have designed a two-branch model DFENet suitable for hazy scenes. One branch extracts the features of the input hazy scene, while the other branch performs a dehazing operation on the input image, using a pretrained dehazing module. We feed the features of the haze branch into the HPM module to predict the f'_{mask}, which will be combined with the features of the two branches and sent to the CFM module for fusion. The fused features guided by the f'_{mask} will enable the detection model to have better performance in hazy scenes while maintaining good performance in clean scenes.

[14], but using generative models often introduces additional noise and artifacts, leading to a decrease in the effectiveness of the model. Some models view haze as a type of noise and prioritize improving the model's noise reduction ability. In addition, there are some methods for studying domain adaptation. Domain adaptation methods are widely used in autonomous driving and object tracking directions, which require models to have high robustness to cope with various weather. Clean images are treated as the source domain, and degraded images containing haze are treated as the target domain. Leveraging the disparities between the source and target domain datasets, the process of model transfer or adaptation is facilitated, which enhances the detection accuracy within the target domain. A prevalent approach in this context entails the simultaneous input of images from diverse domains. By employing gradient inversion layers to obscure the outputs of domain classifiers, the network is enabled to discern features that are invariant to domain variations. This strategy fosters the extraction of robust features that transcend the idiosyncrasies of individual domains, thereby bolstering the generalizability of the model across varied environmental conditions. There is still a very serious problem with object detection models under common hazy conditions, as the model does not know whether the input images come from hazy conditions. When clean images enter the dehazing module, it will bring noise to the clean images and instead lead to a decrease in the performance of subsequent advanced visual task models.

Currently, there are several popular datasets containing rotation annotations, and DOTA [15] mainly focuses on aerial object detection and tracking tasks, aiming to provide large-scale aerial image data, where objects appear at different angles and postures. However, this dataset is not particularly focused on hazy scenarios. HRSC [16] is a dataset focused on ship object detection, which includes ship instances from satellite and aerial images. The ship images in the dataset cover different directions and rotation angles. We noticed that most of the images in these datasets were clear images taken under sunny conditions, neglecting the learning of degraded image features due to weather effects. So more datasets containing degraded remote sensing images are needed to verify the robustness of the model.

To this end, as illustrated in Fig. 2, the structure of DFENet we propose is a two-branch structure, where the input image is fed into the dehazing module to extract the features of the hazy scene image. We use the haze-predict module (HPM) to extract haze information and predict masks, while the cross-fuse module (CFM) combines the features of two branches with the haze mask to obtain the fused features. The use of haze mask fused features can better solve the problem of performance degradation caused by clean input images in the inference stage of the model. In order to have better indicators to prove that our proposed model has a better performance compared to traditional networks, we propose a dataset HRSI, hazy remote sensing images in the real world, and demonstrate that our module outperforms traditional models in HRSI. The main contributions are summarized as follows.

1) A remote sensing image dataset HRSI containing most of the degraded images during hazy weather was proposed, which uses rotated borders for annotation, making remote sensing detection more user-friendly and adding new experimental indicators to the field of remote sensing object detection.
2) We propose a detection model DFENet, a dehazing feature enhancement model for hazy remote sensing images. We conducted tests on HRSI, DOTA, and Foggy-DOTA [17], and compared with traditional mainstream methods, our method outperforms other methods in the field of hazy remote sensing detection.
3) Our model can dynamically balance the weight of the input image and the dehazing image in the feature during the inference stage, reducing the impact of the dehazing module on the detection model.

II. Related Work

A. Oriented Object Detection

For remote sensing object detection, most methods currently rely on predicting frames with angles. SCRDet [18] increases adaptation to small remote sensing objects through improved

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smooth L_1 loss and a network structure with finer sampling capabilities. Zhang et al. [19] designed a context-aware network for learning global and local context relationships in remote sensing images. Xie et al. [9] improved the RPN layer and ROI layer on the basis of Faster R-CNN [3], and improved the regression method of the detector. Adding adaptive points to the network and designing adaptive point learning strategies have enabled more flexible directed object detection. RSDet [20] improved the regression method by directly regressing the four points of the rotation box, and added a correction strategy to address the discontinuity problem in the loss function. Han et al. [7] used a new alignment convolution in the network to generate high-quality anchor and alignment features. On the other hand, in order to address the possibility of large deviation values in angle regression, [21] chose to use angle prediction as a classification task. Yang et al. [18], Qian et al. [22], and Yang et al. [23] were an improvement in the loss function to further optimize the model performance.

B. Object Detection in Hazy Weather

Traditional object detection frameworks frequently demonstrate diminished robustness when confronted with complex backgrounds, occlusion, and variations in object scale. This limitation arises from their reliance on handcrafted features and sliding window techniques for object search, which are inherently less adaptable to the variability in object appearances and environmental shifts. Furthermore, these models are particularly vulnerable to inaccuracies when object boundaries are indistinct, resulting in a decline in detection precision. The challenge is exacerbated in the context of degraded images with diminished quality, where the model struggles to accurately extract and discern the underlying features.

Image restoration is the main task, and these methods mainly divide object detection in hazy weather into two tasks, namely, first removing haze and then detecting, and detecting the faded image. Li et al. [11] proposed an image dehazing model constructed using convolutional neural networks, which is designed based on a redesigned atmospheric scattering model. By introducing adaptive optimal estimation theory into a deep learning framework, the adaptability and optimality of image denoising are achieved. Dong et al. [12] proposed a multiscale enhanced dehazing network with dense feature fusion based on the U-Net architecture. He et al. [24] proposed a very simple prior method, which directly uses the dark channel of the input image to obtain atmospheric light, and then uses the atmospheric scattering model for dehezing. This method does not require training and has a small computational complexity. Dai et al. [25] and Pei et al. [26] believed that the strategy of first removing haze and then detecting would reduce the detection accuracy of the model in most cases. Sun et al. [27] believed that generating models often lose information and generate error patches during the image generation process. A significant challenge in the domain of image processing is the separation of dehazing and detection tasks, which complicates model optimization. This separation can lead to disjointed training stages, ultimately yielding end-to-end models with subpar performance. To address this, researchers have proposed a fine-tuning pipeline dedicated to image restoration, complemented by a momentum-based framework that aims to bolster the training process. However, the scarcity of real-world datasets containing both hazy and clear images poses a considerable obstacle. The prevalent practice in dehazing model training involves artificially introducing haze into clear images, a method that often fails to replicate the complexities of real hazy conditions, thereby impeding the model's ability to generalize effectively to actual hazy scenarios.

Many methods solve problems from the perspective of domain adaptation. Li et al. [28] proposed an adversarial gradient flipping layer and a domain level metric regularization, which can ensure that the domain level metric constraints between the source domain, auxiliary domain, and target domain are regularized during learning, enhancing the learning of difficult sample information. Hnewa and Radha [29] proposed a backbone network with multiscale feature inputs and multiple domain classifiers, which can enable the model to learn invariant features at different levels of features [30] proposed a new decoupling method based on vector decomposition, designing an extractor to separate domain invariant information from input and introducing domain-specific representations as the difference between input and domain invariant representations. By using differential operations, the gap between the invariant information in the target domain and the source domain has been widened. Shan et al. [31] transform the image from the source domain to the target domain and then train the model using adversarial learning. There are also some other methods, such as using multiple filters to denoise and enhance images, and the parameters of the filters can be obtained through model learning. Li et al. [32] proposed a two-branch structure that combines clean and degraded images, and designed an iterative training strategy. Jiao et al. [33] regard haze as a type of noise, and train adversarial networks by adding auxiliary noise to improve the model's defense capabilities.

C. Dataset

Currently, several widely used datasets containing degraded images are being used, and [34] proposes a large-scale dataset consisting of synthetic and real-world blurred images. Generally used for single image dehazing algorithms and some object detection models [35]. Based on cityscpces, an improved atmospheric scattering model algorithm was added with artificial haze, with some of the images coming from the real hazy scene. DOTA [15] is a dataset specifically designed for the field of remote sensing, consisting of 15 categories, mainly consisting of aerial images and top-down angle images. Foggy-DOTA [17] is a simulated hazy scene image synthesized using the imgag library based on DOTA.

III. Method

A. Haze Mask

Atmospheric scattering models [36] are often used to quantify the relationship between hazy and clean images, and many methods utilize atmospheric scattering models to generate

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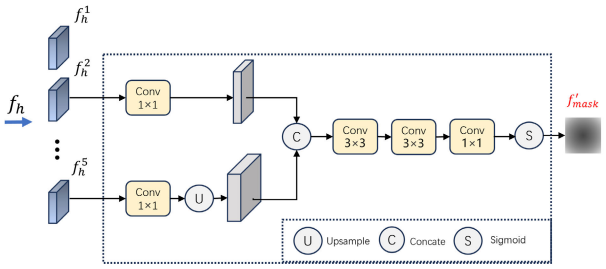


Fig. 3. Our HPM structure. By using this module to extract the features of haze in blurred images, a mask containing information on the location and concentration of haze is predicted.

blurry images, its formula is as follows:

$$I(x) = J(x)t(x) + A(1 - t(x)) \tag{1}$$

where $I(x)$ is the hazy image and $J(x)$ represents the scene radiance (clean image). A is the global atmospheric light and $t(x)$ is the medium transmission map, which is defined as

$$t(x) = e^{-\beta \times d(x)} \tag{2}$$

where β represents the scattering coefficient of the atmosphere and $d(x)$ is the scene depth

$$\beta = \gamma \times \varepsilon, \quad \varepsilon \in [1, 2, 3, \ldots, 9] \tag{3}$$

where γ is the basic scattering coefficient we set and ε is the haze level, used to control the concentration of haze. By controlling different ε values, we can obtain $t(x)$ corresponding to different concentrations of haze

$$G_{\text{mask}} = e^{-\gamma \times \varepsilon \times d(x)}. \tag{4}$$

We obtained haze masks G_{mask} corresponding to different concentrations of hazy weather scenes from (4). Based on (4), we obtain haze mask and composite hazy image from Algorithm 1, and these haze masks will be used in subsequent modules.

B. Haze-Predict Module

Based on the atmospheric scattering model, hazy images can be seen as clean images fused with haze layers. We want to learn haze information through the model, including haze location information and haze concentration information. First, the blurred image is processed through a backbone network and FPN layer to obtain multilayer features $f_{\text{haze}_i}(i = 1, 2, 3, 4, 5)$, i represents different levels. As shown in Fig. 3, we select f_{haze_2} and f_{haze_5}, which contain low-level semantic features and high-level semantic features of the image, respectively. We will send f_{haze_2} and f_{haze_5} together into the HPM module, as shown in Fig. 3, f_{haze_2} and f_{haze_5} each pass through a convolutional layer, and then upsample f_{haze_5} to obtain f'_{haze_5}. We concatenate f_{haze_2} and f'_{haze_5} in the channel dimension to obtain f_k. Finally, we perform three convolution operations on the obtained f_k and activate with sigmoid to obtain f'_{mask}.

C. Cross-Fuse Module

We designed an attention module to receive deep features obtained from two branches. When the features obtained by the dehazing module lose a significant amount of details from the original image, the performance of the model will suffer a serious decline. For this purpose, we have designed a cross fuels module that gives lower weights to areas with dense haze. At this point, the fused features will rely more on the features of clean images. We hope to balance the relationship between clear and blurred images with weighted attention. As shown in Fig. 4, we will use the f'_{mask} obtained from HPM as our haze attention, and in the training stage we use the real label G_{mask} instead of the f'_{mask}. First, we fuse the two features obtained from the backbone network, its formula is as follows:

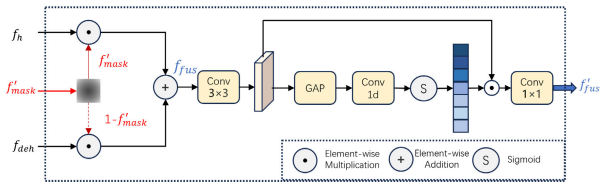


Fig. 4. Illustration of the cross-cilizes the mask predicted by the HPM module to guide the fusion of image features in hazy and clean conditions.

$$f_{\text{fus}} = f'_{\text{mask}} \odot f_{\text{haze}} + \left(1 - f'_{\text{mask}}\right) \odot f_{\text{dehaze}} \tag{5}$$

where f'_{mask} is guided by the haze mask obtained by the adding haze algorithm, which can fully reflect the distribution of haze concentration. When the haze concentration is higher, f_{fus} is more dependent on f_{dehaze}, and when the haze concentration is lower, f_{fus} is more dependent on f_{haze}. We attempt to dynamically integrate f_{haze} and f_{dehaze} in this way so that the model can avoid the performance degradation caused by noise from the dehazing module even in the absence of haze in the input image. f_{fus} will send to a 3×3 convolution to obtain the initial fusion features f'_{fus}, We will obtain f'_{fus} through the following formula:

$$f'_{\text{fus}} = \text{Conv}_{3\times3}(\delta(\text{Conv}_{1\text{D}}(\text{GAP}(f_{\text{fus}}))) \odot f_{\text{fus}}) \tag{6}$$

where $\text{Conv}_{1\text{D}}$ is the 1-D convolution, $\odot$ is the element-wise multiplication, and δ denotes the Sigmoid activation function. Through a channel attention module, we can further reduce redundant channels and suppress noise in features. f'_{fus} will be fed into the detector for regression and classification.

D. Dynamic Balance Mechanism

We obtained a G_{mask} in Algorithm 1, which is a medium transmission map that can indirectly reflect the distribution of haze added by the algorithm and ε represents the depth of haze added to the current image, and the larger ε, the smaller the weight of the center position of the G_{mask}. During the training process, each input hazed image corresponds to a unique G_{mask}. In Fig. 5, we can see that the weight of G_{mask} gradually radiates outward from the center position. During the training process, we can use G_{mask} to guide the model to learn the distribution of G_{mask}, and during the inference process, the model can predict the f'_{mask} corresponding to the input image through HPM. During the inference process, we balance the feature fusion between the input image and the dehazing image based on the f'_{mask} predicted by the HPM. And synthesize features according to the following formula:

$$f_{\text{fus}} = f_{\text{input}} \odot f'_{\text{mask}} + D\left(f_{\text{input}}\right) \odot \left(1 - f'_{\text{mask}}\right) \tag{7}$$

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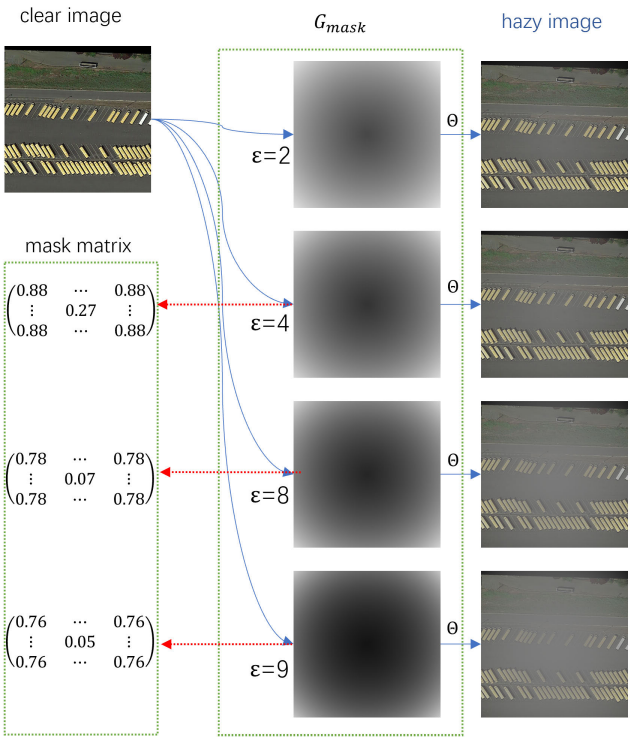


Fig. 5. Illustration of adding haze process. The blue solid line points to the process of adding haze, while the red dashed line points to the weight matrix corresponding to the weight G_{mask}. We can intuitively see that G_{mask} radiates outward from the center, with smaller values near the center of the matrix and larger values away from the center. The ε is haze level, as the haze level ε increases, the center of the G_{mask} becomes smaller and smaller, Θ is the composite function defined in Algorithm 1.

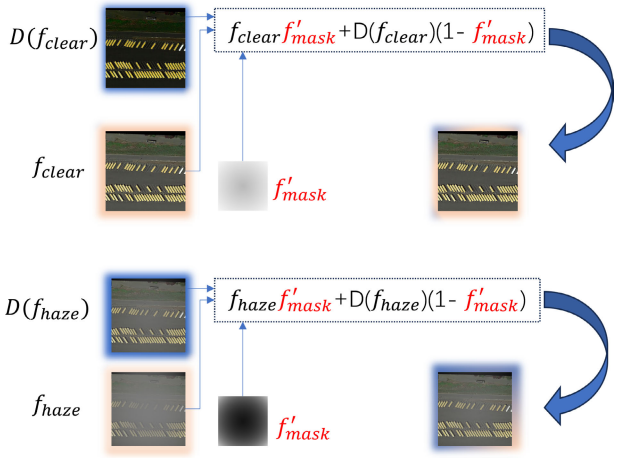


Fig. 6. Illustration of dynamic balance mechanism. In the inference stage, when dealing with hazy and clean image models, the weight size will be dynamically adjusted based on the f'_{mask} to make the synthesized features more inclined toward clean features.

where D is the dehazing module and f_{fus} represents the fused features of the input image features and the dehazing image features. In Fig. 6, we can see that the fusion feature f_{fus} obtained when inputting hazy images will result in smaller weights of f'_{mask} predicted by the HPM, and f_{fus} will be more inclined toward the features of the image after dehazing. When the input image is a clean image, the model predicts a larger weight of f'_{mask}, and f_{fus} tends to be more inclined toward the features of the clean image itself, thereby reducing the impact of features after dehazing the clean image.

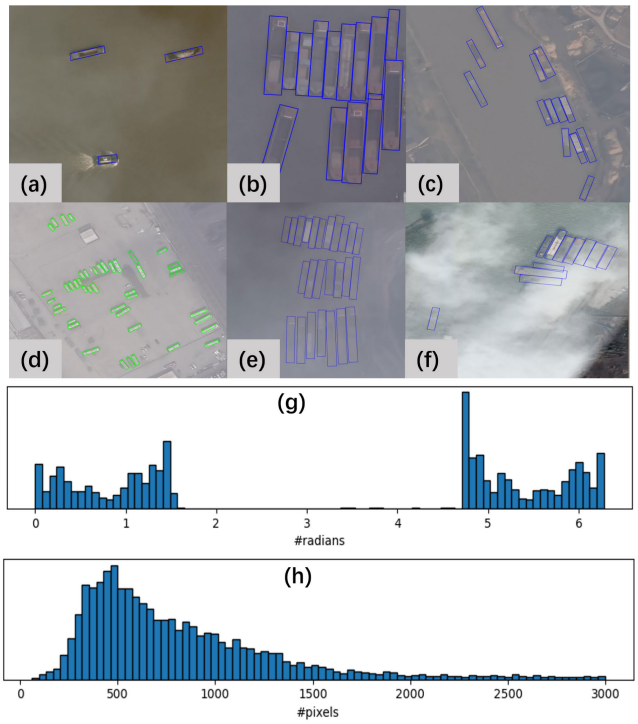


Fig. 7. Display detailed information of HRSI. (a) Example of light fog in HRSI. (b)–(d) Examples of medium fog concentration in HRSI. (e) and (f) Scene with high fog concentration in HRSI. (g) and (h) Display the orientation distribution and size of instances in HRSI. The blue box represents the ship, and the green box represents the large vehicle.

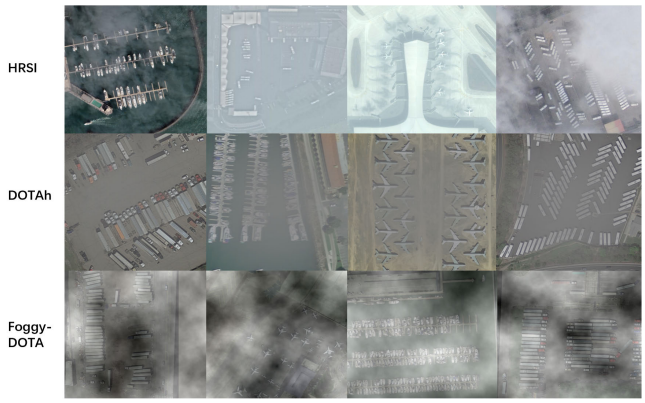


Fig. 8. Sample of the HRSI that we proposed. The first row is our proposed HRSI, and all images are from real hazy scenes. The second row is the DOTAh synthesized on DOTA based on Algorithm 1 in this experiment. The third row is the Foggy-DOTA, which is composed of hazy images based on the imgag library.

E. Training Strategy

We add haze of random concentration to clean image c through Algorithm 1, and by entering the parameters brightness A and basic scattering coefficient γ, we can get haze image m and the corresponding G_{mask}, and the depth of the hazing of each picture is controlled by ε. We use m and c to train the model and G_{mask} to guide the HPM module to predict f'_{mask}. We use FCOS [1] as

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Algorithm 1 Procedure of Data Augmentation by Adding Haze on Clean Image and Generating Haze Mask

Input: Clear image c, Brightness A, basic scattering coefficient γ.
Output: Generated hazy image m, haze mask G_{mask}.

$(H, W, _) = c.shape$;
Define $\Theta(x, y) = x * y + A * (1 - y)$;
$\varepsilon = random(1, 10)$;
$\beta = \gamma * \varepsilon$;
$mask = numpy.zeros(H, W)$;
foreach for j in range(W) **do**
  foreach for k in range(H) **do**
    $d = -0.04 * \sqrt{(j - W//2)^2 + (k - H//2)^2} + \sqrt{\max(H, W)}$;
    $mask[i][j] = math.exp(-\beta * d)$;
$G_{mask} = mask$;
$m = \Theta(c, G_{mask})$;

Algorithm 2 Training and Inferencing Processes of CFM Module and HPM Module

Input: Hazy feature f_{haze}, dehaze feature f_{dehaze}, training status is_train.
Output: Fuse feature f'_{fus}, haze prediction loss L_{mse}

Define HPM, CFM represent haze-predict module and cross-fuse module, respectively.
Define MSE represents the Mean Squared Error loss.
if is_train **then**
  $f'_{mask} = HPM(f_{haze})$;
  $L_{mse} = MSE(f'_{mask}, G_{mask})$;
  $f'_{fus} = CFM(f_{haze}, f_{dehaze}, G_{mask})$;
else
  $f'_{mask} = HPM(f_{haze})$;
  $f'_{fus} = CFM(f_{haze}, f_{dehaze}, f'_{mask})$;

our detection module. We have modified the structure of the backbone to make the entire model a two-branch structure. The input images will go through the dehazing module to obtain the dehazed images, and then the original and dehazed images will be sent to the backbone network to obtain advanced semantic information. We have retained the original detection header and loss function of FCOS.

F. Loss Function

We will not make any changes to the loss of classification and regression in the detector. For the f'_{mask} output from the HPM module, we select the corresponding G_{mask} based on ε in the hazing Algorithm used for the model input image. In Algorithm 1, we can obtain each input hazed image and its corresponding G_{mask}. We use mean squared error loss to guide the HPM module to converge, and the specific formula is as follows:

$$L_{\text{mse}} = \text{MSE}\left(f'_{\text{mask}}, G_{\text{mask}}\right). \tag{8}$$

TABLE I
DETAILED COMPOSITION OF THE DATASET USED IN THIS EXPERIMENT

	HRSI (ours)	DOTAn-tv (DOTAh-tv)	DOTAn-test (DOTAh-test)	Foggy-DOTA (Thin,Thick,Random)
large-vehicle	118541	16580	4364	4364
ship	6219	14224	8677	8667
plane	496	6244	2012	2012
images	787	592	198	198
Total	25256	37048	15053	15053

Finally, we obtained the total loss

$$L_{\text{total}} = wL_{\text{mse}} + L_{\text{reg}} + L_{\text{class}} \tag{9}$$

where w is a hyperparameter used to balance the total loss, and w is set to 0.1 in this experiment. The L_{reg} and L_{class} distributions represent classification loss and regression loss from the detector.

G. Hazy Remote Sensing Dataset

To the best of our knowledge, there are currently few publicly available datasets for hazy remote sensing images in the real world. For this purpose, we propose a dataset HRSI with a total of 796 sheets ranging in size from 512 × 512 to 4000 × 4000 images, including images of various satellite magnifications, directions, and shapes. All images are blurry due to degraded hazy conditions. For particularly blurry images, we have multiple annotators make a joint judgment. We mainly annotated three categories in the HRSI dataset, large vehicle, ship, and plane. These three categories are the main categories of transportation. For the aircraft category, we mainly selected passenger planes in airports. For the ship category, we mainly selected cargo ships and mail ships in canals. For large vehicles, we mainly selected trucks, buses, and trucks. We mainly collect images from bus stations, docks, rivers, and airports. The pixel-to-real map scale of all images is 93:0.71 (pixels: meters) and 93:1.78 (pixels: meters). After browsing through a large number of remote sensing images, we conducted strict screening, mainly selecting degraded images caused by hazy scenes, with most of the objects being blurry or obscured by haze. For objects that are severely degraded and difficult to distinguish manually, we have appropriately discarded some of them. For ambiguous objects, multiple staff members will jointly annotate them. In addition, Fig. 7 showcased other details of the HRSI.

In terms of annotation boxes, we chose the same OBB annotation as DOTA, OBB regression has five values (x, y, w, h, a), and HBB regression has four values (x, y, w, h), where x and y represent the center point position of the border, w and h represent the length and width of the border, and a represents the angle of border rotation. OBB has one more angle information than regular HBB, the shapes of airplanes, ships, and trucks are mainly regular rectangles, and the objects in remote sensing images have high-density characteristics. Directly using nonangled borders will result in a larger overlap of the borders. Therefore, we use angled borders like DOTA for annotation, which has been proven to be superior to HBB-type annotation boxes in the field of remote sensing images, and more suitable for object detection tasks in the remote sensing direction. The specific composition

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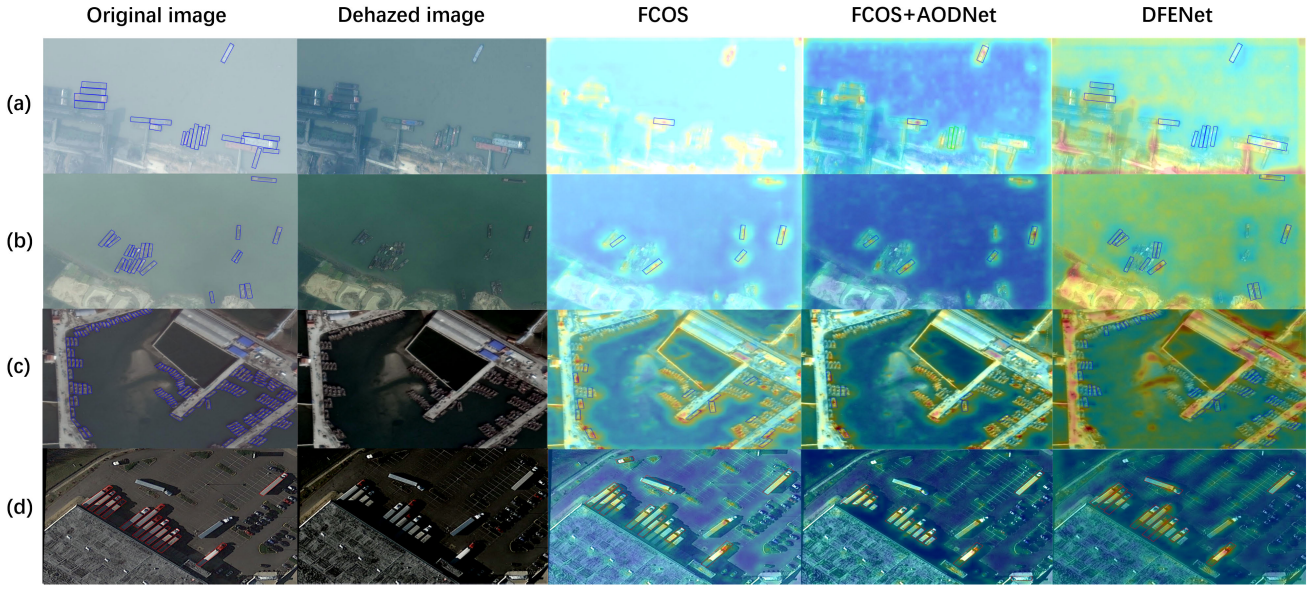


Fig. 9. Detection visualization results from different methods. (a) and (b) Hazy images. (c) and (d) Clear images. When the input image is a clean image, the defogging module will cause a decline in the model's performance. It can be seen that our method brings significant improvements for subsequent detection. The red box represents the large vehicle and the blue box represents the ship.

of the HRSI dataset can be found in Table I and Fig. 8 sample of the HRSI that we proposed.

IV. Experiments

A. Dataset

1) HRSI: HRSI is the dataset we proposed in this experiment, which mainly includes three categories: large vehicle, ship, and plane. Most of the images are selected due to degradation caused by hazy weather. We use HRSI as an indicator that our proposed model can cope with hazy weather conditions. we will first crop images with unequal resolutions to 1024 × 1024, with a cropping window overlap area of 20% pixels in the current image, for sizes smaller than 1024 × 1024 We will enlarge the image to 1024 × 1024 size.

2) DOTAn: DOTAn mainly selects from DOTA [15], DOTA is a large-scale dataset used to evaluate the detection performance of models in remote sensing images, consisting of 2806 images, 15 categories, and various directions, scales, shapes and includes high-density objects. Image size from 800 × 800 to 4000 × 4000 different. In our experiment, We will only use the training set and validation set from DOTA, and we will use the validation set for testing. In simple terms, DOTAh is entirely based on DOTA, and the division of the training set and validation set is also completely consistent with DOTA, all images from DOTAn-tv come from the training set of DOTA, and all images from DOTAn-test come from the validation set of DOTA, for a detailed distribution, refer to Table I, which include images from three categories: large vehicle, ship, and plane. Other images and object borders of other categories will be filtered out. Similarly, all images will be cropped to 1024 with a sliding window of 10% overlapping pixels 1024 × 1024 size. The specific division of the dataset is shown in Table I.

3) DOTAh: DOTAh is all obtained by adding haze based on the atmospheric scattering model on the basis of DOTAn. We randomly added haze of different concentrations on the basis of DOTAn according to the method in Algorithm 1.

4) Foggy-DOTA: Foggy-DOTA is a dataset synthesized using the imgag library based on the DOTAn dataset. In this experiment, it includes three subdatasets: Thin Foggy-DOTA, Thick Foggy-DOTA, and Random Foggy-DOTA. The images in Thick Foggy-DOTA contain high concentrations and wide distribution of synthetic haze, while Thick Foggy-DOTA contains shallow synthetic haze. Random Foggy-DOTA is obtained by randomly mixing images of high-concentration synthetic haze with images of low-concentration synthetic haze. In this experiment, Foggy-DOTA will be used as the validation set to evaluate the performance of the model.

B. Implementation Details

All of our experiments are deployed on a server with NVIDIA Geforce GTX 3080ti GPU and 12 G memory. In the preprocessing section before training, we only selected one random flipping enhancement method. The entire model uses an SGD optimizer. The initial learning rate is set to 0.0025, the weight decay is 10^{-4}. We trained 12 epochs on the DOTAh-tv dataset, and the learning rate decreased by 10% in the eighth and ninth epochs.

C. Experimental Results

We chose the FCOS with the Resnet50-FPN as the baseline and divided the experimental strategies into the following three categories.

1) *Direct:* Input images containing hazy scenes directly into the model for training.
2) *Separation:* We first train the detection model on VOCn-tv, then train the dehazing module, and input the images into the dehazing module for detection.
3) *Union:* The detection model utilizes a dehazing module during the training phase. To ensure the accuracy of the

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TABLE II
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON DOTA AND HRSI

Strategy	Methods	Module		Train Dataset	Test Dataset		
		Dehazing	Detection		DOTAn-test	DOTAh-test	HRSI
Direct	baseline-1	/	FCOS [1]	DOTAn-tv	82.11	71.36	48.01
	baseline-2	/	FCOS	DOTAh-tv	82.41	82.73	52.05
Separate	Sep-1	AODNet [11]	FCOS	DOTAn-tv,DOTAh-tv	80.28	75.22	47.79
	Sep-2	MSBDN [12]	FCOS	DOTAn-tv,DOTAh-tv	80.99	77.85	51.08
	Sep-3	GridDehaze [13]	FCOS	DOTAn-tv,DOTAh-tv	81.00	77.69	50.08
Union	IA-YOLO [37]	/	Faster R-CNN [3]	DOTAh-tv	80.95	81.22	50.91
	IA-YOLO	/	FCOS	DOTAh-tv	79.49	80.02	51.43
	DS-Net [38]	AODNet	FCOS	DOTAn-tv,DOTAh-tv	81.76	82.51	52.17
	CascadeNet [39]	AODNet	FCOS	DOTAn-tv,DOTAh-tv	69.53	73.11	43.48
	BAD-Net [32]	AODNet	FCOS	DOTAh-tv	85.60	86.61	56.44
	DFENet(ours)	AODNet	FCOS	DOTAh-tv	**86.74**	**86.87**	**58.30**

experiment, we changed the detection module in other models to the baseline for quantitative comparison.

IA-YOLO designs filters based on dark channels prior to dehazing, adding a branch to the model to predict the parameters of the filter. CascadeNet mainly includes a cascading training strategy, pretraining a detection model, fixing the detection model parameters, and embedding the dehazing module into the detection model. The entire training process only updates the parameters of the dehazing module. DS-Net adds a recovery branch behind the backbone network and uses recovery loss to perform dehazing operations on the input images during the training phase. BAD-Net proposed a two-branch structure that combines clean and degraded images, unlike DFENet, it directly concatenates the fused features with the dual-branch features. As shown in Table II, we compared the average accuracy of the total three types of target (mAP) results between existing technical methods. There are four main observations.

1) The results of the direct training strategy indicate that using only clean images to train the model will lead to a decrease in performance in both synthetic haze (DOTAh) and real haze (HRSI) scenes. Adding synthetic haze to the training set can help improve the robustness of the model.
2) The separated training strategy can improve the model to a certain extent in hazy scenes, but using this strategy for clean images will introduce noise, leading to a decrease in the mAP of the model on DOTAn. However, among different dehazing modules, AODNet will lead to a slight decrease in model performance in hazy scenes, while MSBDN and Grid will improve.
3) CascadeNet has poor performance and is difficult to converge, as the convergence of the model during the training process only depends on the dehazing module. At the same time, without updating the model parameters, it is difficult for the model to learn new features. IA-YOLO is learning that hazy scenes mainly rely on dehazing filters. However, dehazing filters use dark channel priors that overly rely on image priors, and the filter has too few parameters to learn, making it difficult for the model to converge. The performance of DS-Net has slightly improved, as the learning of the recovery branch during the training process makes it easier for the backbone network to distinguish hazy scenes. The BAD-Net performs well, thanks to its dual-branch structure, where the dehazing features supplement the original features, enabling the model to capture more information from degraded images.
4) DFENet is a two-branch model that also includes a prediction module for hazing masks on images. By using masks, we can more easily learn features on branches with lower noise during training. In the inference stage, the model can also make appropriate tradeoffs between the features of two branches based on the predicted mask. Fig. 9 is detection visualization results from different methods, it show that our model has a dynamic balance mechanism for two branches. When the input image is a clean image, the model tends to use the features of the clean image itself. When the input image contains haze, the model tends to use the features of the dehazing image. In addition, Fig. 10 shows the visualization of our model's detection results on different datasets. We conducted multiple experiments and presented the results in Fig. 11.

D. Ablation Study

1) Effectiveness of Each Module in the Model: Perform ablation studies to analyze the performance of each module in DFENet. We will evaluate the branch structure, loss function, CFM module, and HPM module of the model. The experimental results are shown in Table III. Compared to a single branch, a two-branch structure leads to a slight improvement in model performance, the two-branch approach directly fuses f_{haze} and f_{dehaze} on the channel dimension, and then feeds them into the detector. On the basis of the two-branch, we tested the CFM module separately. Since there was no HPM predicted f'_{mask} as guidance, we directly fuse f_{haze} and f_{dehaze} on the channel dimension. When testing the performance of the CFM module without f'_{mask} as guidance, we found that the performance of the model improved by 3.87 mAP compared to the two-branch structure on the HRSI dataset.

We use both CFM and HPM modules on the basis of two branches, and the HPM module provides f'_{mask} for the CFM module. We found a significant improvement in model performance. When using L_1 loss as the f'_{mask} prediction loss, we improved the HRSI dataset by 0.51 mAP compared to using only the CFM module. In addition, both the DOTAn-test test and the DOTAh-test test showed improvements. When we

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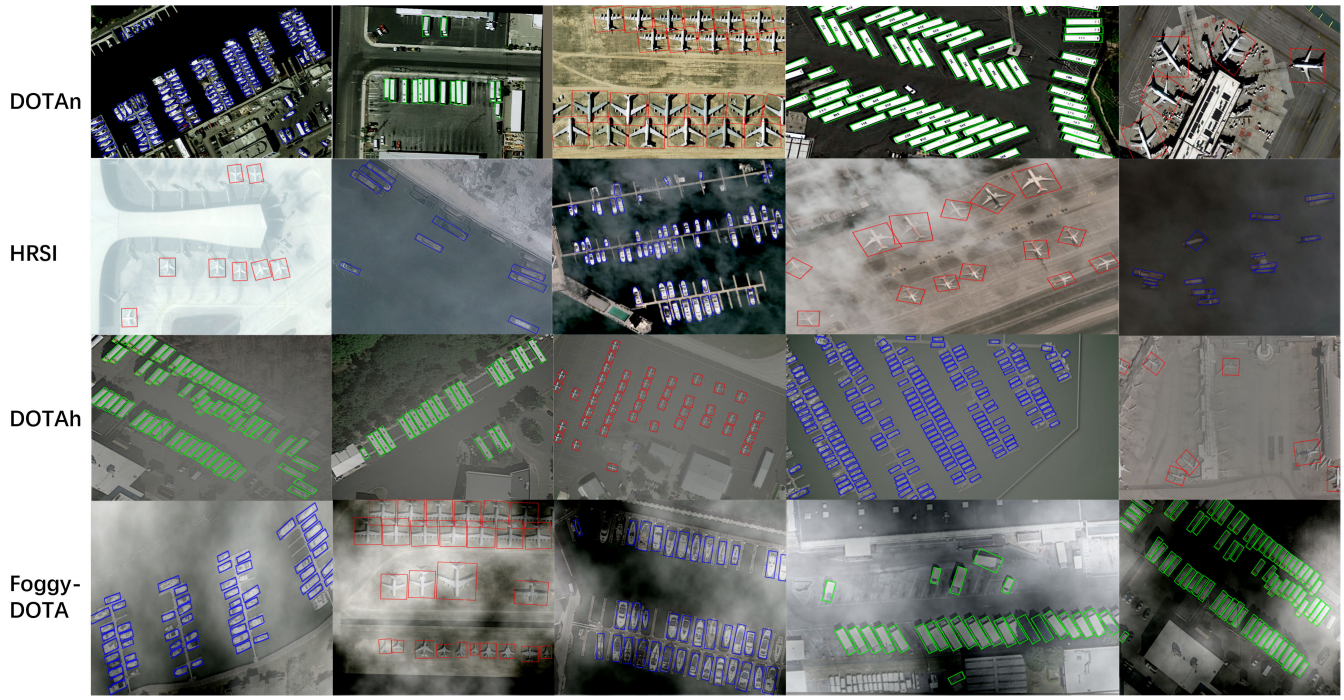


Fig. 10. Example detection results of DFENet on different datasets. The first row is the DOTAn-test, the second row is our proposed HRSI, and all images are from real hazy scenes. The third row is the DOTAh-test synthesized on DOTA based on Algorithm 1 in this experiment. The fourth row is the Foggy-DOTA, which is composed of hazy images based on the imgag library. The red box represents the plane, the blue box represents the ship, and the green box represents the large vehicle.

TABLE III
PERFORMANCE OF DIFFERENT MODULE COMBINATIONS ON DFENET

Two Branch	CFM	HPM	L_1	L_2	DOTAn-test	DOTAh-test	HRSI
					82.41	82.73	52.05
✓					83.20	83.34	52.83
✓	✓				86.26	86.11	56.70
✓	✓	✓	✓		86.50	86.48	57.21
✓	✓	✓		✓	**86.74**	**86.87**	**58.30**

TABLE IV
COMPARISON WITH OTHER UNIVERSAL DETECTORS

Method	DOTAn-test	DOTAh-test	HRSI
RetinaNet [2]	78.90	79.31	48.91
FCOS [1]	82.41	82.73	52.05
R3Det [10]	75.41	75.72	48.28
S2A-Net [7]	82.82	82.78	52.57
ReDet [40]	**87.65**	**87.94**	51.75
Oriented R-CNN [9]	86.96	87.11	52.76
DFENet(ours)	86.74	86.87	**58.30**

TABLE V
PERFORMANCE TESTING OF OUR MODEL ON FOGGY-DOTA

Method	Train Dataset	Foggy-DOTA		
		Thin	Thick	Random
FCOS(baseline)	DOTAn-tv*	72.88	61.57	65.19
FCOS(baseline)	DOTAh-tv	82.10	79.74	80.01
RetinaNet	DOTAh-tv	83.60	80.36	82.09
R3Det	DOTAh-tv	81.03	79.81	80.74
S2A-Net	DOTAh-tv	82.12	81.77	81.89
Oriented R-CNN	DOTAh-tv	87.01	82.51	85.28
ReDet	DOTAh-tv	**87.99**	85.49	**87.76**
DFENet(ours)	DOTAh-tv	86.60	**85.59**	86.16

use L_2 loss, the performance improvement of the model is the highest, with an improvement of 6.25 mAP on HRSI. At the same time, it improves by 4.33 and 4.41 mAP on DOTAn-test and DOTAh-test compared with baseline, respectively. Experimental results show that these modules have a significant improvement in the performance of the model.

2) Comparison With Traditional Object Detection Methods on HRSI: Our model compares traditional object detection models, including universal object detectors and detectors in the remote sensing field. The training set uses DOTAh-tv, and for quantitative analysis, all models use Resnet50-FPN as the backbone network. Among them, RetinaNet [2] and FCOS [1] are universal target detectors. S2A-Net [7], R3Det [10], Oriented R-CNN [9], and ReDet [40] are detectors that have been improved for the field of remote sensing. In Table IV, we can see that Oriented R-CNN and ReDet have good performance in both DOTAh-test and DOTAn-test. This is because they are two-stage detectors that generate a large number of candidate boxes in the first stage, covering more possible target positions. However, their performance is poor on the HRSI dataset, and the model's generalization and robustness are poor. RetinaNet, FCOS, S2A-Net, and R3Det are one-stage detectors, and experimental results show that the performance of two-stage detectors is better than that of one-stage detectors. Our model has the best performance on HRSI, and also performs well on DOTAh-test and DOTAn-test, outperforming most detectors.

3) Testing on the Synthetic Dataset Foggy-DOTA: We tested our model and other models on Foggy-DOTA, which includes three subdatasets. The results are shown in Table V. We trained the model using DOTAn-tv and DOTAh-tv and found that using DOTAh-tv as the training set significantly improved the testing of Foggy-DOTA. Because DOTAh-tv and Foggy-DOTA

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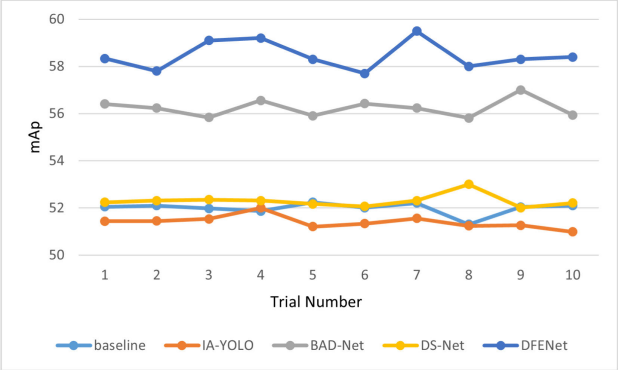


Fig. 11. Display of results from multiple experiments. We conducted multiple experiments on HRSI with various models and plotted the results of each experiment, following the same configuration as in Table II. Our DFENet outperformed other models in the mAP distribution across multiple experiments. The vertical axis represents the mAP tested on HRSI, and the horizontal axis represents the number of experiments.

TABLE VI
USING DIFFERENT DEHAZING MODULES IN OUR MODEL

Model	DOTAn-test	DOTAh-test	HRSI
AODNet	**86.74**	**86.87**	58.30
MSBDN	86.70	86.78	**58.47**
GridDehaze	86.73	86.71	57.79

TABLE VII
PERFORMANCE OF DIFFERENT DETECTION MODULES

Model	DOTAn-test	DOTAh-test	HRSI
FCOS(baseline)	**86.74**	**86.78**	**58.47**
Oriented R-CNN	84.01	85.59	56.10
S2A-Net	81.49	81.11	55.13
R3Det	78.15	78.69	51.13

are both DOTA plus artificially synthesized haze, they have a common distribution. In the testing of other models, we found that the two-stage model Oriented R-CNN and ReDet are better than the one-stage model, and have good performance on all three subsets of Foggy-DOTA. Our proposed HRSI is better than the one-stage models R3Det, S2A-Net, and RetinaNet. At the same time, our model performs better than Oriented R-CNN on the subdatasets Thick and Random, indicating that our model has good performance under dense haze conditions.

4) Impact of Different Dehazing Modules on the Model: We used the same DFENet configuration as in Table II, and replaced the dehazing module, the results are shown in Table VI. Overall, the impact of different dehazing modules on the model was relatively small. MSBDN has more learnable parameters, and its performance has slightly improved compared to AODNet.

5) Performance of Different Detection Modules: We replaced the detection module in DFENet with the following common detectors: R3Det, S2A-Net, Oriented R-CNN, and tested them on HRSI. As shown in Table VII We found that the detection module has a significant impact on the performance of DFENet. Overall, R3Det and S2A-Net are single-stage models with lower performance than two-stage models Oriented R-CNN. Although FCOS is a single-stage model, it has a higher number of parameters and better performance in terms of speed and performance.

TABLE VIII
EFFICIENCY COMPARISON WITH DIFFERENT MODELS ON HRSI

Model	Params	Speed(ms)	mAP
FCOS(baseline)	31.89M	**38**	52.05
Oriented R-CNN	41.3M	43	52.76
DS-Net(FCOS)	31.91M	68	52.17
DFENet(ours)	45.16M	79	**58.30**

6) Efficiency Analysis: The parameter quantity and speed of the model are shown in Table VIII. All models use Resnet50-FPN as the backbone network, and we tested HRSI on NVIDIA 3080 GPU. All images have a resolution of 1080 × 1080. Compared with other models, DFENet has an increased number of parameters and inference time. Our model has a good improvement in detection accuracy while ensuring real-time performance under the condition of frame extraction.

V. CONCLUSION

This article proposes a degraded dataset for hazy remote sensing images, mainly targeting hazy weather scenarios. This dataset contains several common remote sensing objects. In addition, we also propose an oriented object detector DFENet, which learns more information through the use of HPM and cross-fuse model. A large number of experiments have shown the effectiveness of our method, and future generations can also use our dataset to continue exploring the field of hazy remote sensing object detection.

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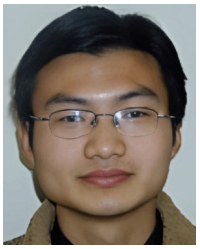
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